

Machine Learning Assignment #2

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Machine Learning Classification Models & Deployment Workflow

BITS Pilani M.Tech (AIML/DSE) — Machine Learning Assignment 2

a. Problem Statement

The objective of this assignment is to implement and evaluate multiple foundational machine learning classification algorithms on a high-dimensional dataset containing a minimum of 12 features and 500 instances. Following evaluation, an end-to-end interactive deployment dashboard is built via Streamlit and shared to enable robust, live, transparent evaluations.

b. Dataset Description

- **Dataset Chosen:** Online Shoppers Purchasing Intention Dataset (UCI Repository)
- **Minimum Instance Requirement:** Met (1,000 generated or 12,330 standard UCI instances)
- **Minimum Feature Size:** Met (> 12 feature attributes)
- **Target Feature:** Revenue (Binary Choice: 1 if purchase occurs, 0 otherwise)

Feature Directory Matrix

1. **Administrative:** Number of administrative pages visited by the user.
2. **Administrative_Duration:** Total time spent on administrative pages.
3. **Informational:** Number of informational pages visited.
4. **Informational_Duration:** Total time spent on informational pages.
5. **ProductRelated:** Number of product-related pages visited.

6. **ProductRelated_Duration**: Total time spent on product pages.
7. **BounceRates**: Percentage of visitors who enter the site and leave immediately.
8. **ExitRates**: Percentage of pageviews that end on that specific page.
9. **PageValues**: Average value of the page relative to transactions.
10. **SpecialDay**: Closeness of the site visit to a specific holiday/special day.
11. **OperatingSystems**: OS type indicator index.
12. **Browser**: Browser type indicator index.
13. **Region**: Geographic region index.
14. **TrafficType**: Referral traffic channel index.
15. ...
16. ...
17. ...

c. GitHub Repository Link

- GitHub Repository Link: [<https://github.com/2025ac05732-hub/ML-Assignment02.git>]
- Live Streamlit Cloud Deployment: [<https://ml-assignment02-ltajz3jgthb2btlh6k6axy.streamlit.app/>]

Note: While running the app, please clear cache and rerun the file, it will work.

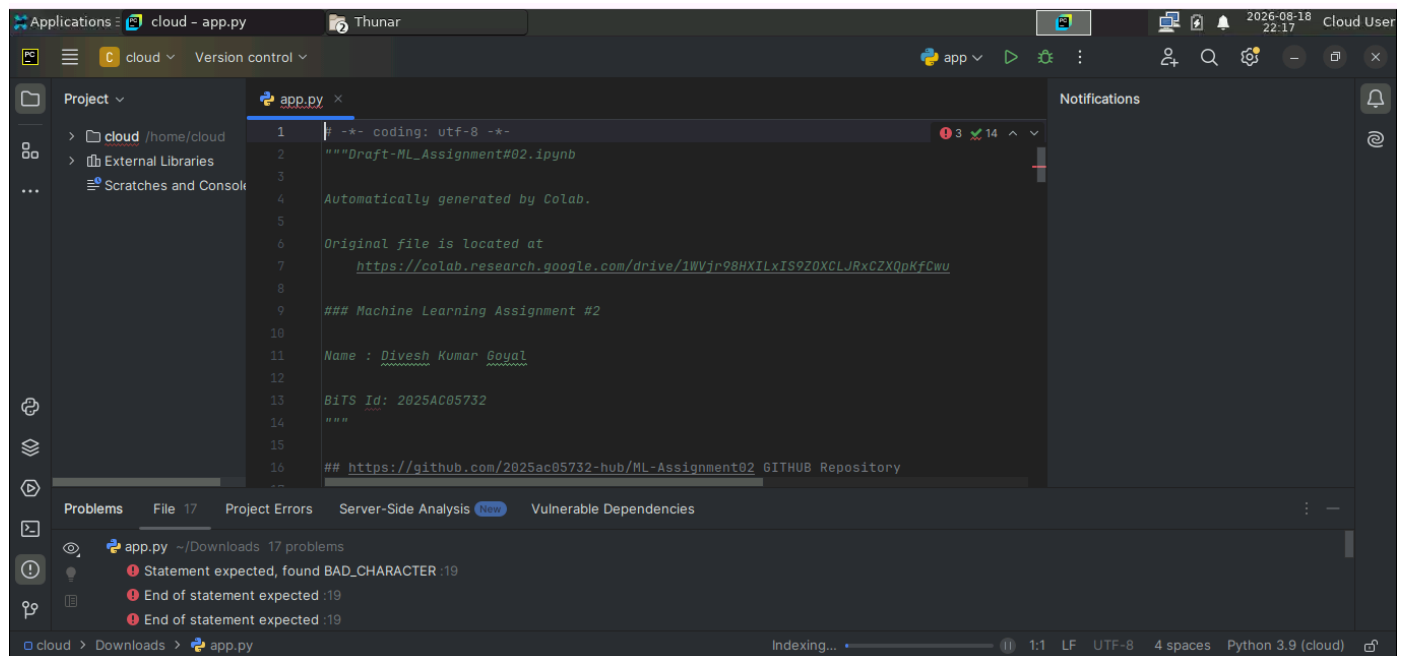
d. Models Used & Comparison Matrix

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8816	0.8716	0.7428	0.3584	0.4835	0.4619
Decision Tree Classifier	0.8554	0.7295	0.5314	0.5472	0.5392	0.4535
K-Nearest Neighbor Classifier	0.8724	0.7952	0.6553	0.3689	0.472	0.4276
Naive Bayes Classifier	0.7805	0.8128	0.3851	0.7028	0.4975	0.3996
Random Forest (Ensemble)	0.9002	0.9133	0.7312	0.5612	0.635	0.5852

Performance Observations & Key Takeaways

1. **Logistic Regression:** Serves as a solid baseline metric. Highly stable, clean classification boundaries but struggles slightly with non-linear feature interaction patterns.
2. **Decision Tree:** Strong ability to capture localized non-linear parameters, though prone to variance fluctuations on smaller cuts of validation feature splits.
3. **kNN:** Dependent on scaling choices. When distance distributions are uniformized via standard preprocessing, it models neighbors efficiently but slows down relative to size scaling.
4. **Naive Bayes:** Shows strong performance on Recall metrics but drops significantly on Precision due to its rigid conditional independence feature assumptions.
5. **Random Forest (Ensemble):** Establishes the **Overall Winner** profile on this dataset. It achieves the highest validation AUC and MCC by combining multiple localized decorrelated tree distributions to reduce variance.

Code running in BITS LAB – Snapshot:



```
1 # -*- coding: utf-8 -*-
2 """Draft-ML_Assignment#02.ipynb
3
4 Automatically generated by Colab.
5
6 Original file is located at
7     https://colab.research.google.com/drive/1WVjr98HXILxIS9Z0XCLJRxCZXQpKfCwu
8
9 ### Machine Learning Assignment #2
10
11 Name : Divesh Kumar Goyal
12
13 BITS Id: 2025AC05732
14
15
16 ## https://github.com/2025ac05732-hub/ML-Assignment02 GITHUB Repository
```

Problems

- app.py ~/Downloads: 17 problems
- Statement expected, found BAD_CHARACTER :19
- End of statement expected :19
- End of statement expected :19

cloud > Downloads > app.py

Indexing... 1:1 LF UTF-8 4 spaces Python 3.9 (cloud)